Plasma proteins

Introduction

Blood proteins also termed plasma proteins are the proteins present in blood plasma. Many proteins are found in plasma because they perform their functions in the blood, others are circulating to a tissue and others because they are released from injured cells.

Some functions performed by plasma proteins are:

- Maintenance of oncotic pressure. Albumin accounts for 40-60% of blood proteins thus, it is a major contributor to maintaining the oncotic pressure of plasma and prevents fluid loss to the tissues.

- Transporting of water-insoluble substances, such as fatty acids or steroid hormones.

- Maintenance of acid-base balance.

- Circulating reserve of amino acids for cells, especially albumin.

- Hemostasis, since numerous factors related to coagulation and fibrinolysis are found in plasma.

- Defense, in which immunoglobulins and C-reactive protein are involved.

The concentration of each protein depends on the rate of synthesis, its distribution between the plasma and the extravascular space, and the rate of degradation.

Most plasma proteins are glycoproteins, such as albumin, with 1% carbohydrate in their molecule. Most plasma proteins are synthesized in the liver. An exception are immunoglobulins, synthesized in plasma cells, and various peptide hormones such as insulin, synthesized by endocrine glands. Protein catabolism occurs mainly in the liver. Plasma proteins are also lost by the kidney and the vascular endothelium.

The study of plasma proteins is performed by quantifying the total concentration of proteins in plasma and by quantifying some individual proteins. The separation by electrophoresis and semiquantification of the different protein fractions is also a valuable diagnostic tool.

Total Plasma proteins

Most laboratories routinely measure total proteins in plasma or serum. The total plasma protein concentration is 6.6-8.7 g/dL. The general causes of changes in the total protein concentration are the variation in the aqueous composition of the plasma and changes in the concentration of the most abundant proteins, mainly albumin and immunoglobulins. Hypoproteinemia is more frequent than hyperproteinemia. (Fig 1)

Hypoproteinemia may be due to decreased protein synthesis or excessive protein losses:

- For the adequate synthesis of plasma proteins, a sufficient supply of amino acids in the diet is necessary, so the nutritional status has a decisive influence on their concentration. Therefore, states of malnutrition or intestinal malabsorption cause hypoproteinemia. Since the liver is the main

organ of plasma protein synthesis, severe liver disease can lead to a decrease in protein synthesis and, consequently, to low plasma protein values. Severe immunodeficiency may be accompanied by hypoproteinemia, due to decreased immunoglobulin synthesis.

- Hypoproteinemia can also associate with kidney dysfunction becauses of high protein losses through the kidney. Extensive burns also produce high protein losses, which cause hypoproteinemia.
- Increased blood volume can also cause hypoproteinemia due to hemodilution. This may be due, for example, to fluid retention.

Hyperproteinemia can also be caused by impaired synthesis or hemoconcentration. A decrease in plasma volume causes hemoconcentration, which is reflected in hyperproteinemia. Hyperproteinemia due to dehydration may be due to decreased fluid intake or increased fluid loss, as in the case of severe diarrhea or vomiting. Immunoglobulin elevation occurs, for example, in patients with multiple myeloma where there is exaggerated clonal synthesis. This means than a specific clon of immunoglobulins is increased. Polyclonal increases of immunoglobulins also cause hyperproteinemia.

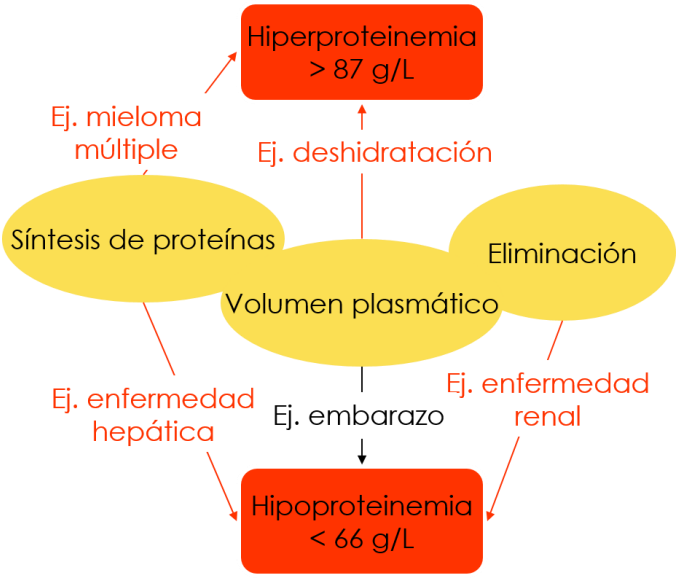


Figure 1. Main causes of alterations in protein concentration due to physiological (pregnancy) or pathological causes.

Protein separation by electrophoresis

Serum proteins are separated by their different electrical charge by agarose gel electrophoresis or by capillary electrophoresis. The electrical charge of the protein depends on the pH of the buffer. A pH of 8.6 is normally used. Immunoglobulins migrate in the opposite direction due to the low charge in relation to electroendosmosis (movement of cations towards the cathode which pulls the surrounding water molecules).

When agarose gel electrophoresis is used, proteins are stained using Coomassie brilliant blue. Electrophoresis separates the proteins into five main bands or fractions (Figure 2): albumin is at the anodic (+) end, followed by the α1-globulin, α2-globulin, β-globulin and γ-globulin fractions. After staining, the gel is visually inspected and the percentage represented by each band is measured by densitometry and the proteinogram profile is obtained.

In capillary electrophoresis, no staining is used and semi-quantification of the bands is performed directly during migration measuring the absorbance of the peptide bond with UV light.

In order to quantify each band, the total protein concentration in the sample is also determined. The proteinogram should be performed with a serum sample, since, with plasma, fibrinogen appears as an additional band in the β-region, similar in appearance to a monoclonal band, which can lead to an error in interpretation.

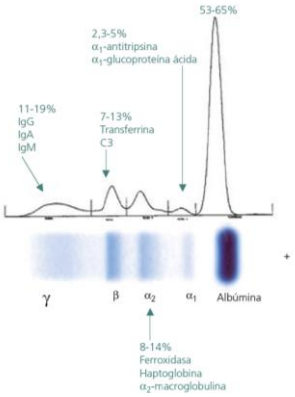


Figure 2. Electrophoresis of serum proteins in alkaline buffer in agarose gel and staining with Coomassie blue. The corresponding densitometric analysis, the proportion of bands and the main proteins influencing each one are shown. Ig: immunoglobulin

Immunofixation

This technique is used to identify a monoclonal protein. Immunofixation has two phases: electrophoresis, in which the components of the sample are separated, and immunoprecipitation of the proteins to be detected.

First, an agarose gel is used in which the sample is applied to each lane and electrophoresis is performed. Subsequently, an antibody that recognizes each of the different immunoglobulin heavy and light chains is added to each lane of the gel. These antibodies diffuse into their assigned lane in the gel and form a precipitate with their corresponding antigen. In a reference lane the proteins are precipitated with a fixative solution. After an incubation time for the immunoprecipitate to form, the gel is washed to remove soluble and non-precipitated proteins, and stained to identify the proteins. (Figure 3)

Incremento policlonal de inmunoglobulinas IgG: 14,30 g/L

Incremento monoclonal de inmunoglobulinas IgG-lambda IgG: 22,80 g/L

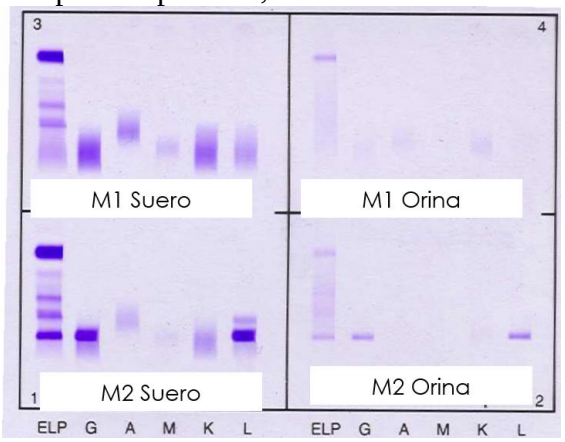


Figure 3. Polyclonal (M1) or monoclonal (M2) increase of immunoglobulins in serum. Example of elimination of light chains in urine. Immunofixation of the serum sample shows the existence of a monoclonal IgG-λ band and free λ light chains. Immunofixation of urine shows elimination of IgG-λ.

Study of the main plasma proteins

Albumin

Albumin is the most abundant protein in plasma, representing between 40 and 60% of the total, making it the main protein responsible for maintaining oncotic pressure. It has a molecular weight of 68.5 kDa and a large amount of negative charges in its molecule. Its half-life is about 21 days.

Albumin is synthesized in the liver according to protein intake and its concentration in blood, and is metabolized in many tissues where it is taken up and hydrolyzed by the cells, taking advantage of its amino acids. In this way, it is a reservoir of amino acids. Catabolism increases in traumatic processes, infections and surgical interventions. Another function of albumin is that of transport, due to its high concentration and the large number of negative charges it possesses. Thus, it transports fatty acids, drugs, vitamins, calcium and bilirubin. It also transports various hormones. Therefore, changes in the plasma concentration of albumin modify the concentration of these substances that it transports.

Hypoalbuminemia is much more common than hyperalbuminemia. The most frequent cause of an increase in its concentration is dehydration. Its concentration can also be raised "artifactually" by venous stasis at the time of extraction. Since albumin is the most abundant protein in plasma, its decrease usually causes hypoproteinemia. Hypoalbuminemia may be due to physiological causes, as occurs during pregnancy, because the circulating volume increases and, to some extent, hemodilution occurs. It can also be due to pathological causes by:

- Lower synthesis, as in malnutrition, where low amino acid intake leads to a decrease in albumin synthesis. For this reason, the determination of albumin in blood is frequently used in the monitoring of the patient's nutritional status, such as in intestinal malabsorption. Albumin concentration also decreases in liver diseases due to a decrease in the synthetic capacity of the liver.

- Losses, such as in renal disorders, can cause very severe hypoalbuminemia. γ-globulinas

γ-globulin

The γ-globulin region of the serum protein electrophoresis consists mainly of immunoglobulins, which are formed by 2 identical heavy chains and 2 identical light chains linked by disulfide bridges (Fig 4). The heavy chains are of 5 types (γ, α, μ, δ and ε) and the light chains are of two types, κ and λ, There are five classes of immunoglobulins, differing in heavy chains: IgG, IgA, IgM, IgD and IgE. Immunoglobulins participate in mechanisms of defense of the organism.

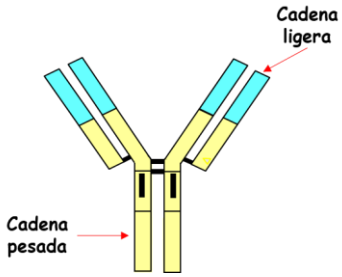


Figure 4. Immunoglobulin structure.

Hypogammaglobulinemia

Immunodeficiencies can be genetic or acquired, and are associated with a risk of recurrent infections. The decrease or absence of the γ-globulin region in the proteinogram usually indicates an IgG deficiency, usually secondary to protein loss or insufficient synthesis.

Hypergammaglobulinemia

The polyclonal increase of immunoglobulins is the normal response to infection, in which all immunoglobulins are elevated. The monoclonal elevation of immunoglobulins is due to the synthesis of molecules with identical structure. If the amount produced is significant, a well-defined band is detected in the γ region of the proteinogram. To type the monoclonal band observed in the proteinogram it is necessary to perform another technique called immunofixation. Immunofixation consists of the use of specific antibodies against the heavy and light chains, so that a band is formed where the monoclonal component of the sample (antigen) reacts with the added antibody.

Multiple myeloma is a malignant neoplasm characterized by the elevation of a clone of plasma cells in the bone marrow, which produce a single type of immunoglobulin, making it monoclonal. At the same time the production of normal immunoglobulins is inhibited, so that these patients frequently present recurrent infections. Typical are osteolytic lesions in bone, with elevated values of alkaline phosphatase and calcium in blood, and other non-specific symptoms such as weight loss, anemia, hemorrhages or renal failure.

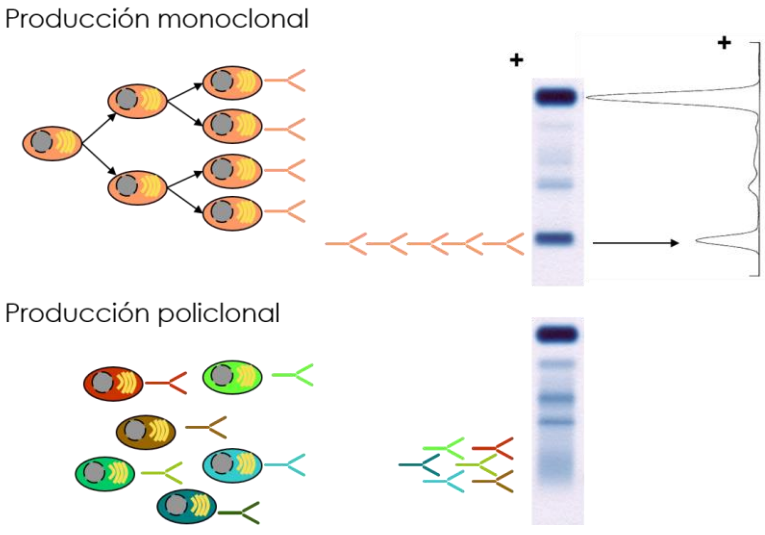


Figure 5. Production of a monoclonal immunoglobulin by a clone of plasma cells against a polyclonal response. Electrophoresis and densitometry of the monoclonal band are indicated.

Acute phase reaction

In the face of an injury of various types, mechanical, physical, chemical, biological or tumor, a defensive response of the organism is activated, called acute phase reaction. In this reaction, certain proteins change their concentration in the blood in response to the inflammation. They are called acute phase reactants:

- Some proteins such as C-reactive protein (CRP) or immunoglobulins increase their concentration. They are positive acute phase reactants.

- Others, such as albumin, decrease in concentration. They are negative acute phase reactants.

Therefore, the ratio between albumin and globulins changes, which leads to alterations in the physicochemical properties of the plasma. Red cells have a negative electrical charge, but the increase in immunoglobulins, with more positive charges, favors red cell sedimentation. Thus, when there is an acute phase reaction, the erythrocyte sedimentation rate increases. It can be measured in the laboratory by the erythrocyte sedimentation rate test, whose elevation in a patient indicates that there is an acute phase reaction.

These proteins have a defensive mission against aggression through different mechanisms. Acute phase reactant proteins are nonspecific, i.e. they are associated with many diseases. Therefore, their determination is not diagnostic, but rather is used to follow the evolution of the disease and the efficacy of treatment.

C-reactive protein

C-reactive protein is synthesized in the liver. It is so named because of its property to bind tightly to the C polysaccharide of the S. pneumoniae cell wall. Upon binding, it activates the classical complement pathway and can initiate processes that promote the elimination of pathogenic cells.

It is the most representative of the acute phase reactant proteins, as its concentration increases dramatically after cell injury. Its elevation is not disease specific, but supports the diagnosis of inflammatory processes. It is very sensitive for diagnosing inflammatory intestinal diseases, and for detecting infections and rejection in transplanted organs. Serial determination of its concentration is of interest, since the levels and the time it remains elevated reflect the severity of the process that originated it and its evolution.

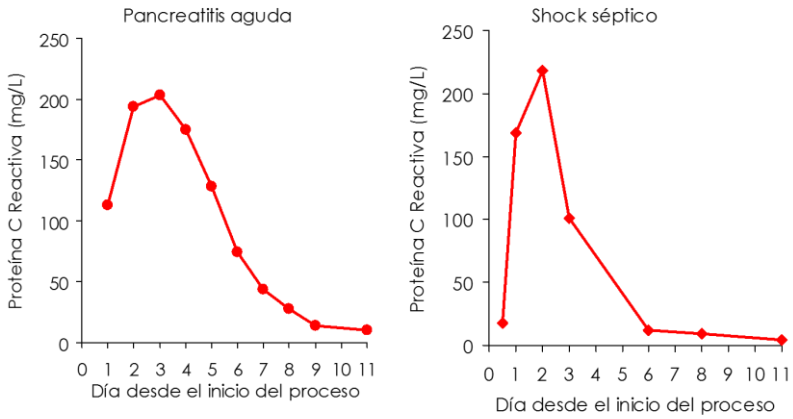


Figure 6. Evolution of C-reactive protein concentration during an acute process: acute pancreatitis and septic shock. Concentrations decrease, reflecting a favorable evolution.